

Project Design Phase - Solution Architecture

Date	Team ID	Project Name	Maximum Marks
18 June 2026		ShopEZ E-Commerce Platform	4 Marks

1. System Architecture Diagram



Syntax error in text
mermaid version 11.15.0

2. Component Descriptions

Frontend (React App)

- **Vite + React SPA:** Renders the user catalog, search input, price selectors, cart preview, and order tracking.
- **AuthContext:** Manages logged-in states, logins, and logouts, syncing `shopez_user` and `shopez_token` to `localStorage`.
- **CartContext:** Reactive state for listing cart quantities and calculating subtotals.
- **Axios Client:** Adds `Authorization: Bearer <token>` automatically and handles global `401 Unauthorized` responses by wiping local storage credentials.

Backend (Express API)

- **Express Web Server:** Exposes endpoints to search products, process transactions, and save orders.
- **Auth Middleware:** Decodes token parameters from requests to ensure secure admin access.
- **Database (MongoDB):** Houses persistent collections for Users, Products, Orders, and Admin records.